Goal

The goal is to flip an array so that the first element becomes the last and the last becomes the first, this should be done for all the elements in the array. Using the GPU this process can be done in parallel.

How To flip?

To flip an array of size N a kernel function is run in parallel enabling multiple of the same operations at once. This means that each element in the array is handled by a different thread. When the number of elements is smaller than the maximum number of threads, one thread block can be used. For larger array sizes, a grid is created containing multiple thread blocks. For example a 2x2 grid contains 4 thread blocks, if each thread block contains 256 threads the total number of threads will be 1024 threads. The grid size and thus number of thread blocks is calculated based on the number of threads per block and number of elements in the array. For each element in the array one thread will copy an element from the input array to the output array at the calculated location using block dimension, block ID, thread ID and number of elements. In order to make the code fully functional, there are a few extra steps. 1: allocate memory on CPU and add value's. 2: allocate memory on the GPU 3: copy data from CPU to GPU memory 4: copy output array from GPU to CPU memory after flipping the array.